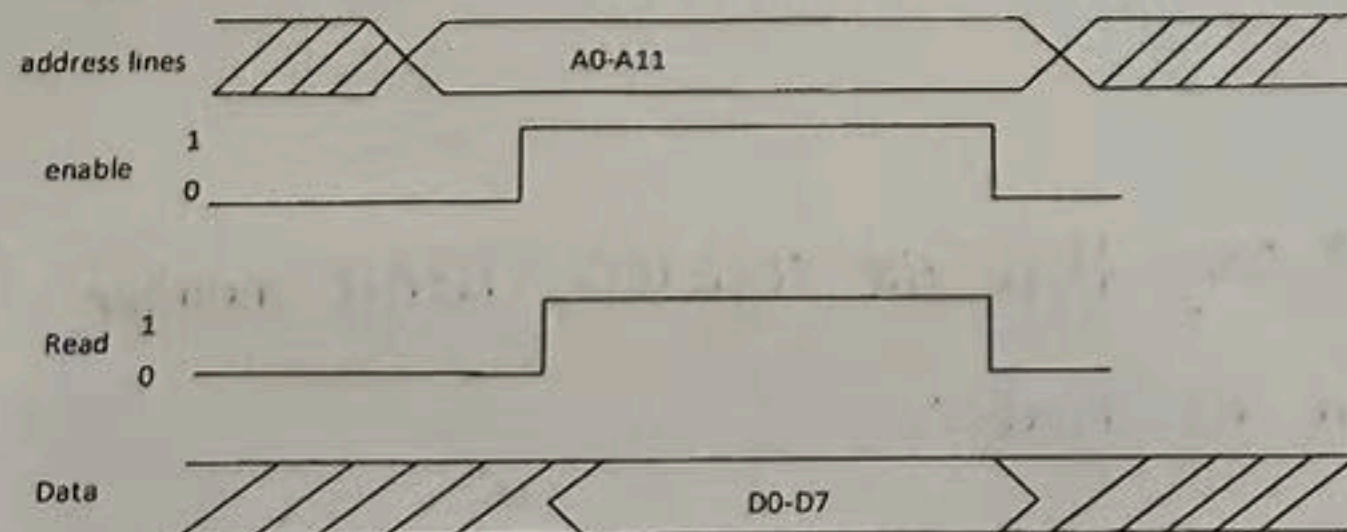


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Quiz 3
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Q1

Figure Q1 is the timing information of a read cycle for data memory of a PIC18 system.



FigureQ1

(a) For Part(a), is the read cycle in the execute phase or in the fetch phase for an instruction cycle? Briefly explain.

(b) The PIC is running the following instruction:

`MOVF 0x1F, Access`

"MOVF 0x1F, Access" means read the content of the location 0x1F using access bank method.

Suppose that before the execution of the instruction, WREG=0xB7 and the value in location 0x1F is 0xAA.

What is the value of data bus and the value of address bus in this read cycle?

(15 marks)

(a) Fetch phase, the ~~cpu~~ is getting data from address bus.

(b) data bus: 0xAA

address bus: 0x1F

Q2

Write a few instructions that initialize WREG to 01H and then move the content of WREG to the RAM location (Data Memory location) 10H.

`MOVLW 0x01`

`MOVWF 0x10, Access`

(5 marks)